

Junior Backend Candidate Exercise

Goal

Build one small package with Melange, then build and test a container image with apko that uses it.

This exercise is meant to simulate the kind of work junior engineers do on the image-building side of the platform. The image is the main artifact. Package work is included because the image must be built from a package you produced yourself.

Tools

- `melange` : a tool for building APK packages from declarative YAML definitions
- `apko` : a tool for building container images from APK packages and declarative image configuration

Suggested Effort

This task would typically be expected to take a few hours to complete.

Exercise

Package and ship the CLI tool `dasel` from upstream version `v3.3.1` .

`dasel` is a small single-binary CLI for querying and modifying structured data formats such as YAML and JSON. For this exercise, use `v3.3.1` , which is affected by `CVE-2026-33320` . As part of the task, make sure the package and resulting image include a fix for that CVE while still building from the `v3.3.1` codebase.

Your task has three parts:

1. Build a Melange package for `dasel v3.3.1` .
2. Build an apko image that installs that package.
3. Prove the final image works with a focused test.

Expected Outputs

Use any clear layout you prefer. A suggested structure is:

- `melange/dasel.yaml`
- `apko/dasel.yaml`
- `tests/test.sh`
- `README.md`

Your submission should include:

- the Melange package definition
- the apko image definition
- a test script that proves the image works
- short notes explaining how to build and test the result
- any additional files needed to implement the `CVE-2026-33320` fix

Packaging Requirements

Requirements:

- fetch or check out the upstream `dasel` `v3.3.1` source
- fix `CVE-2026-33320` while staying on the `v3.3.1` codebase
- build the `dasel` binary with Melange
- include at least one focused package test

Image Requirements

Requirements:

- create an apko image definition for `dasel`
- make the image install the package you built, not a prebuilt upstream image
- keep the image thin and centered on the `dasel` binary

The image test must prove both:

- the expected tool is present and runnable
- one real `dasel` behavior works inside the container

Important Constraint

One of the key parts of this task is connecting the package build to the image build.

Your apko build must consume the package artifact you created locally. You may choose any reasonable local setup for that, as long as your `README.md` explains it clearly.

Build And Test

Use standard local commands for Melange and apko.

Your `README.md` should include the exact commands needed to:

- build the package
- test the package
- build the image
- run the image test

You may use Docker or another reasonable local container runtime for testing.

Success Criteria

The submission is successful if:

- `dasel v3.3.1` is packaged correctly with Melange
- the package includes a fix for `CVE-2026-33320`
- the image is built with apko and installs the package produced by the candidate
- the image can be built successfully on a normal local setup
- the image test demonstrates real runtime behavior, not only a version or help check
- the submission notes explain how to run the solution and mention any assumptions

Submission

Submit your work along with a short note that includes:

- the commands you ran
- whether they passed fully
- anything you would improve with more time